PRODUCT DATA SHEET

DUROTECH DUROCEM PU SL

Polyurethane concrete flooring system

DUROTECH
Engineered Construction Chemicals
Made In Australia For Over 45 Years

DESCRIPTION

Durotech Durocem PU SL is a unique polyurethane concrete product which is applied as a topping on concrete floors where difficult service conditions are the norm. Durotech Durocem PU SL is recommended for use where exposure to temperatures up to 120 °C is expected. For these conditions a minimum of 9 mm is recommended. Durotech Durocem PU SL is also the product of choice where maximum resistance to acetic acid, lactic acid and other organic acids is required. Resistance to food acid (eg phosphoric acid) and sanitizing chemicals such as peroxyacetic acid solutions is also excellent.

The system is based on a solventless polyurethane and a cementitious quartz filler blend. The product is usually applied as a 6 mm broadcast topping complete with a range of anti-slip finishes. This broadcast system should be sealed with Durotech Durocem PU TC.

Durotech Durocem PU SL can also be applied as a single layer, smooth seamless 4 mm self-levelling topping.

Features

- Heavy duty, durable, abrasion and impact resistant
- Chemically resistant to acids, caustics, fats and cleaning solutions
- Fast installation with no odour issues
- Can be applied to 7 day old concrete
- Moisture tolerant and can be applied to a slightly damp surface
- Available with a chemical accelerator for fast cure time even at low temperatures
- Available in a variety of non-slip textures to meet and exceed Australian Standards for safety
- Supplied as a pre-packaged kit for uniform consistency
- Available in a range of colours

TECHNICAL INFORMATION

Mixing ratio by weight - corresponds to packaging in kg	
A : B **A : B : C**	3 : 3 3 : 3 : 14
Pot life @ 25 °C	approx. 20 minutes
Application temperature (min. 3 °C above dew point)	10 to +30 °C
Material consumption	1.9 kg/m2 per mm
Overcoating @ 25 °C	12 -24 hours
Cure time @ 25 °C: **Foot traffic** **heavy traffic** **chemical exposure**	 12 - 20 hours 2 days 7 days
Coefficient of expansion - Durotech Durocem PU SL	14 x 10-6 $°C^{-1}$
Coefficient of expansion - Concrete	12 x 10-6 $°C^{-1}$
Adhesive strength	> 1.5 MPa
Tensile strength	7 MPa
Elastic Modulus	2 GPa
Flexural strength	15 - 20 MPa
Compressive strength	45 - 55 MPa
Temperature resistance	-20 °C to 120 °C
Hardness (Shore D)	80 (7 days)
Coverage	2.5sqm/kit

SINCE 1977

INDUSTRY APPLICATIONS

- Abattoirs and meat processing establishments
- Bakeries and snack food manufacturers
- The fish processing industry
- Breweries, wineries, fruit juice and soft drink manufacturers
- Cold storage warehouses, cool rooms and freezers
- The dairy industry including milk, butter, cheese and yoghurt products
- The edible oil and margarine industry
- The sugar industry including refineries and confectionery producer. The chemical and mining industry, containment bunds in mineral processing and treatment plants
- Ammonium Nitrate storage
- Areas subject to constant heavy traffic and chemicals such as tanker bays, waste transfer stations, fertilizer plants, large machinery workshops, bulk liquids transfer etc.
- Commercial and industrial kitchens in restaurants, hospitals, catering facilities, etc.
- Bars, food retailers, supermarket food preparation areas

CHEMICAL RESISTANCE

CHEMICAL	CHEMICAL
Organic acids	
Acetic acid 5%	Excellent
Acetic acid 10%	Excellent
Acetic acid 20%	Good
Butyric acid 10%	Excellent
Citric acid 50%	Excellent
Lactic acid 15%	Excellent
Oleic acid	Excellent
Mineral acids	
Concentrated Hydrochloric	Good
Nitric acid 5%	Excellent
Nitric acid 20%	Good
Nitric acid 35%	Spillage and 8 hour cleanup
Phosphoric acid 15%	Excellent
Phosphoric acid 35%	Excellent
Phosphoric acid conc.	Spillage and 8 hour cleanup

CHEMICAL	CHEMICAL
Fats, oils & solvents	
Animal fats	Excellent
Ethyl alcohol	Good
Kerosine	Excellent
Lubricating oils	Excellent
Aromatic and Ketone Solvents	Poor to fair
Petrol unleaded	Excellent
Skydrol	Good
Vegetable oils	Excellent
Water-based chemicals	
Ammonium hydroxide 20%	Excellent
Ferric chloride 10%	Excellent
Salt solutions 10%	Excellent
Sodium hyperchloride 16%	Good
Sodium hydroxide 20%	Excellent

SUBSTRATE PREPARATION

The concrete substrate should be firm, clean and dry. The compressive strength of the surface must be a minimum of 25 MPa and the surface tensile strength a minimum of 1.5 Mpa.

New concrete is to be a minimum of 7 days old. The surface of the concrete will need to be mechanically prepared by either diamond grinding, shot blasting or scarifying. All weak and loose material, surface laitance, contaminants and coatings or curing compounds must be completely removed. It is preferred that the prepared surface has a profile to ensure both excellent chemical and mechanical adhesion.

Anchoring grooves minimum 10 mm wide and 10 mm deep will need to be cut just inside the perimeter of the area to be topped as well as around drains. Double diamond blade saw cuts 6 x 6 mm must be placed over the entire floor at maximum distance of 4 m intervals.

Porous concrete will require priming, if unsure then onsite tests must be carried out.

PRIMING

Apply a 1mm scratch coat of Durotech Durocem PU SL to the prepared surface.

APPLICATION GUIDELINES

SAFETY PRECAUTIONS

Wear gloves, eye protection masks and overalls during mixing and application. Refer to msds for further information.

MIXING INSTRUCTIONS

The resin and hardener are to be mixed together in the correct ratios using a mechanical variable speed helical mixer for at least 1 - 2 minutes prior to the addition of the bag of Durotech Durocem PU SL aggregate. Continue mixing until all the ingredients have been incorporated and are thoroughly mixed, with the final mix being totally homogenous. It is important to maintain continuity of mixed material on site to maintain the wet edge of the applied topping.

SELF LEVELLING 4MM TOPPING

Apply the mixed product over the prepared surface using either pin rake or notched trowel set at the correct thickness (3-4mm). The topping is then rolled with a spiked roller to remove entrained and entrapped air.

ANTI-SLIP 6MM TOPPING

Where a 6 mm anti-slip topping is required it is necessary to broadcast the applied 4 mm Self levelling topping with the chosen aggregate to a beach finish and allowed to cure. Once cured the excess aggregate is swept and vacuumed away. The surface is now ready for sealing with Durotech Durocem PU TC. The non-slip texture of the system will be determined by the size of the aggregate used.

ANIT-SLIP 9MM TOPPING

Apply the Durotech Durocem PU SL broadcast system in two layers to achieve a final film thickness of 9 mm using the Durotech Durocem PU TC as a final seal coat.

Aggregates

The hardness of the broadcasting aggregate will impact on the durability and lifespan of the applied flooring system. The harder the aggregate, the better the long term performance. The following list shows a comparative hardness of some commonly available aggregates.

Quartz	Sand Hard
Bauxite	Harder
Aluminium Oxide	Hardest

For heavy duty environments where exposure to regular hard wheeled traffic or constant mechanical abrasion is common, Durotech strongly recommend consideration be given to choosing either Bauxite or Aluminium Oxide as the broadcasting aggregate.

COVERAGE

2.5sqm/kit

PACKAGING

Durotech Durocem PU SL is a 3 component product consisting of a resin, hardener and blended fillers.

Component breakdown:

Part A	3 kg
Part B	3 kg
Part C Aggregate	14 kg

This mix yields 10.50 Litres

LIMITATIONS

Durotech Durocem PU SL is an industrial flooring finish which may discolour on exposure to UV light from the sun or an artifical source. The severity of discolouration is dependant on colour choice. Any such discolouration has no effect on the performance of the product.

SHELF LIFE

This product has a shelf life of 9 months from date of manufacture, stored away from sunlight at 25 °C in original un-opened container.

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